Actividad 5

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Fundamentación de robótica (Gpo 101)

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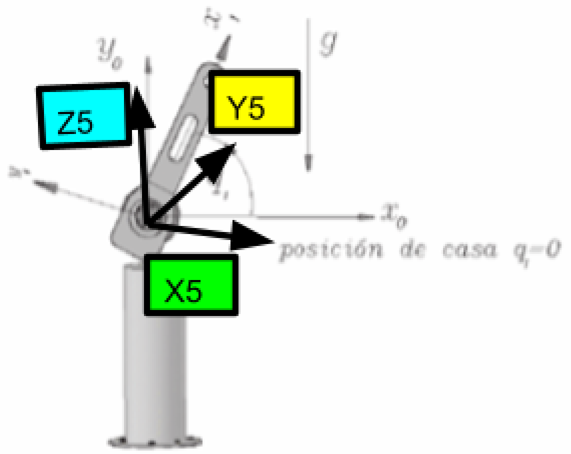


Figura 4.10 Péndulo robot.

Robot Péndulo (1gdl)

Rotación: No hay rotación

$$\begin{vmatrix} \cos(\theta_1) & -\mathrm{sen}(\theta_1) & 0 \\ \mathrm{sen}(\theta_1) & \cos(\theta_1) & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

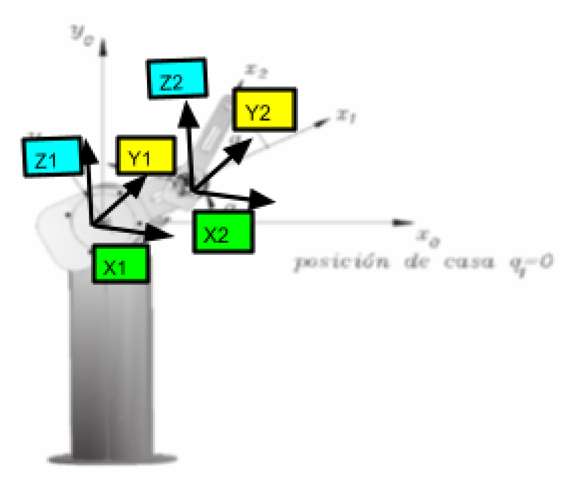
Posición:

$$\begin{vmatrix} l1*\cos(\Theta\_1) \\ l1*\mathrm{sen}(\Theta\_1) \\ 0 \end{vmatrix}$$

```
Energia cinetica del pendulo
     2         /  d           \2
(m lc  + Izz) | -- theta(t) |
               \ dt          /
------------------------------
              2



Lagrangiano
/      2         /  d          \2                             \
| (m lc  + Izz) | -- theta(t) |                               |
|                \ dt          /                              |
| ------------------------------ - g lc m cos(theta(2)) |
|               2                                             |
|                                                             |
|      2         /  d          \2                             |
| (m lc  + Izz) | -- theta(t) |                               |
|                \ dt          /                              |
| ------------------------------ - g lc m sin(theta(2)) |
|               2                                             |
|                                                             |
|                 2          /  d          \2                  |
|          (m lc  + Izz) | -- theta(t) |                       |
|                         \ dt          /                      |
|          ------------------------------                      |
\                        2                                    /
```



Robot Rotacional (2gdl)

R^0_1 :

$$\begin{vmatrix} \cos(\theta_1) & -\mathrm{sen}(\theta_1) & 0 \\ \mathrm{sen}(\theta_1) & \cos(\theta_1) & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

R^{1}_{2} :

$$
\begin{vmatrix}
\cos(\theta_1) & -\text{sen}(\theta_1) & 0 \\
\text{sen}(\theta_1) & \cos(\theta_1) & 0 \\
0 & 0 & 1
\end{vmatrix}
$$

Posición 1:

$$
\begin{vmatrix}
l1*\cos(\Theta\_1) \\
l1*\text{sen}(\Theta\_1) \\
0
\end{vmatrix}
$$

Posición 2:

$$
\begin{vmatrix}
l1*\cos(\Theta\_2) + l2*\cos(\Theta\_1+\Theta\_2) \\
l1*\text{sen}(\Theta\_2) + l2*\text{sen}(\Theta\_1+\Theta\_2) \\
0
\end{vmatrix}
$$

```
Energía Cinética Total (Robot 2 GDL):

       2                                                               2           2            2
(m1 lc1  + Izz1) #1   Izz2 #1   Izz2 #2         d          d          l1  m2 #1   lc2  m2 #1   lc2  m2 #2      2    d          d                                                               d          d
------------------ + ------- + ------- + Izz2 -- th1(t) -- th2(t) + --------- + ---------- + ---------- + lc2  m2 -- th1(t) -- th2(t) + l1 lc2 m2 cos(th2(t)) #1 + l1 lc2 m2 cos(th2(t)) -- th1(t) -- th2
        2                 2         2          dt         dt            2            2            2               dt         dt                                                              dt         dt

where

         /  d        \2
   #1 == | -- th1(t) |
         \ dt        /

         /  d        \2
   #2 == | -- th2(t) |
         \ dt        /
```

```
Lagrangiano:

/      Izz2 #11   Izz2 #10                                                                                                 \
| #5 + -------- + -------- + #3 - g m2 (lc2 cos(#4) + l1 cos(th1(2))) + #9 + #8 + #7 + #2 - g lc1 m1 cos(th1(2)) + #6 + #1 |
|         2          2                                                                                                     |
|                                                                                                                          |
|      Izz2 #11   Izz2 #10                                                                                                 |
| #5 + -------- + -------- + #3 + #9 + #8 + #7 - g m2 (lc2 sin(#4) + l1 sin(th1(2))) + #2 - g lc1 m1 sin(th1(2)) + #6 + #1 |
|         2          2                                                                                                     |
|                                                                                                                          |
|                                        Izz2 #11   Izz2 #10                                                               |
|                                   #5 + -------- + -------- + #3 + #9 + #8 + #7 + #2 + #6 + #1                            |
\                                          2          2                                                                    /

where

                                  d          d
   #1 == l1 lc2 m2 cos(th2(t)) -- th1(t) -- th2(t)
                                 dt         dt
```

```
          2      d          d
#2 == lc2  m2 -- th1(t) -- th2(t)
                 dt         dt

              d          d
#3 == Izz2 -- th1(t) -- th2(t)
             dt         dt

#4 == th1(2) + th2(2)

                2
      (m1 lc1  + Izz1) #11
#5 == --------------------
                   2

#6 == l1 lc2 m2 cos(th2(t)) #11
```

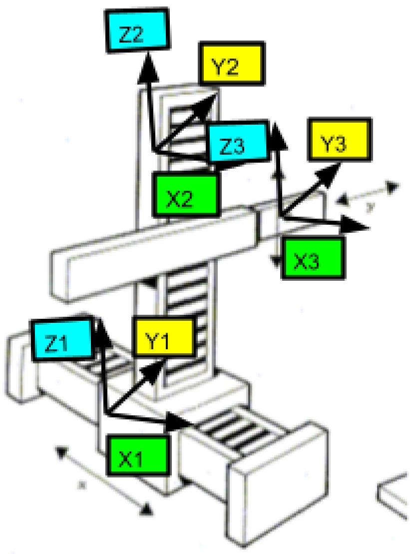
```
          2
       lc2  m2 #10
#7 == ----------
            2

          2
       lc2  m2 #11
#8 == ----------
            2

         2
       l1  m2 #11
#9 == ---------
            2
```

```
         /  d        \2
#10 == | -- th2(t) |
         \ dt        /

         /  d        \2
#11 == | -- th1(t) |
         \ dt        /
```



Robot Cartesiano (3gdl)

No hay rotación:

$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

P1:

$$\begin{vmatrix} q1 \\ 0 \\ 0 \end{vmatrix}$$

P2:

$$\begin{vmatrix} 0 \\ q2 \\ 0 \end{vmatrix}$$

P3:

$$\begin{vmatrix} 0 \\ 0 \\ q3 \end{vmatrix}$$

```
Energia cinetica robot cartesiano

   /                 /  d      \2 \                        /                                /  d       \2 \
   |                 | -- y(t) |  |                        |                                | -- z(t)  |  |
   | /  d       \2   \ dt       /  |      /  d      \2      | /  d       \2   /  d      \2   \ dt       /  |
m2 | | -- x(t)  |  + ------------ |   m1 | -- x(t) |    m3 | | -- x(t)  |  + | -- y(t) |  + ------------  |
   \ \ dt       /          4      /      \ dt       /       \ \ dt       /    \ dt       /          4       /
----------------------------------- + --------------- + ---------------------------------------------------
                 2                           8                                   2

Lagrangiano

/         /  d      \2                  \
|      m1 | -- x(t) |                   |
|         \ dt      /                   |
| #2 + --------------- + #1 - g m3 x(3) |
|             8                         |
|                                       |
|         /  d      \2                  |
|      m1 | -- x(t) |                   |
|         \ dt      /                   |
| #2 + --------------- + #1 - g m3 y(3) |
|             8                         |
|                                       |
|         /  d      \2                  |
|      m1 | -- x(t) |                   |
|         \ dt      /         g m3 z(3) |
| #2 + --------------- + #1 - --------- |
\             8                    2    /
```